Integrated Circuit ECE326

Lec06

Synthesis Digital Circuit

Dr. Ahmed Reda Mohamed[1]

[1]PhD, the MoE Key Laboratory of Artificial Intelligence & Bio-Circuits and Systems Laboratory (*BiCASL*), Department of Micro-Nano Electronics, Shanghai Jiao Tong University, Shanghai, China.

Outlines

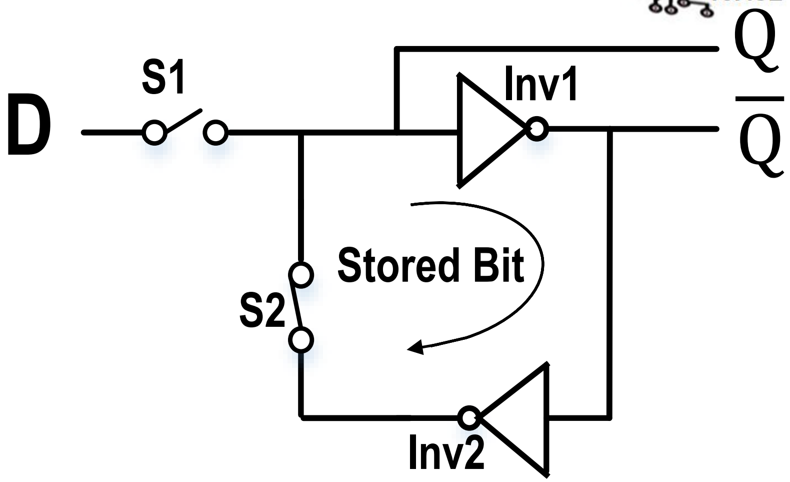
- Memory Cell
- D-Latch
- D-Flip-Flop

Outlines

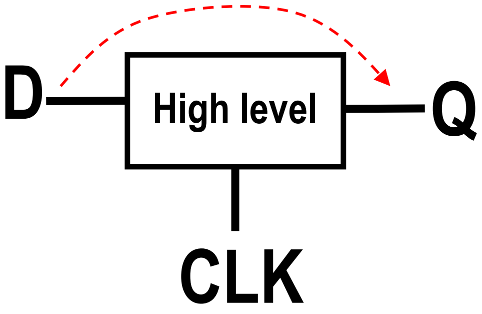
- ❑ **Memory Cell**
- ❑ **D-Latch**
- ❑ **D-Flip-Flop**

Memory Cell (1/2)

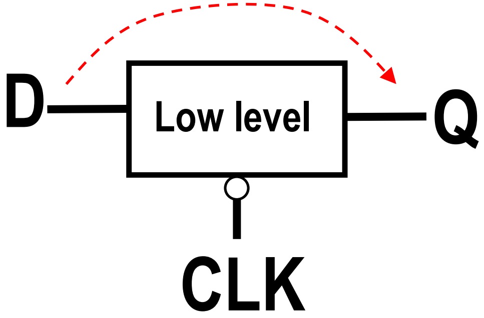
- **Definition**
  - Memory cell is an element which is used to store single bit.
- **Construction**
  - Consists of two inverters and two switches as shown.



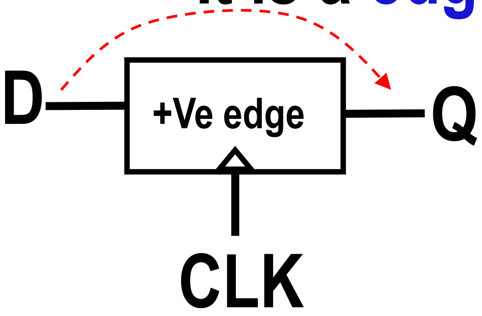
D-Latch	D- Flip Flop
It is a level sensitive	It is a edge sensitive



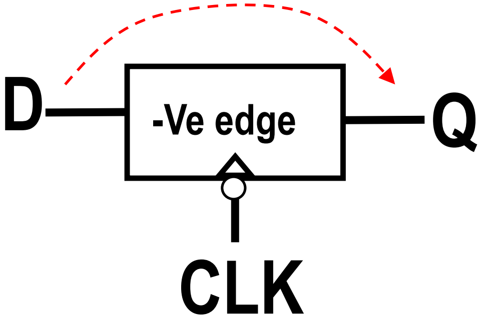
@CLK= “1”, Q=D



@CLK= “0”, Q=D



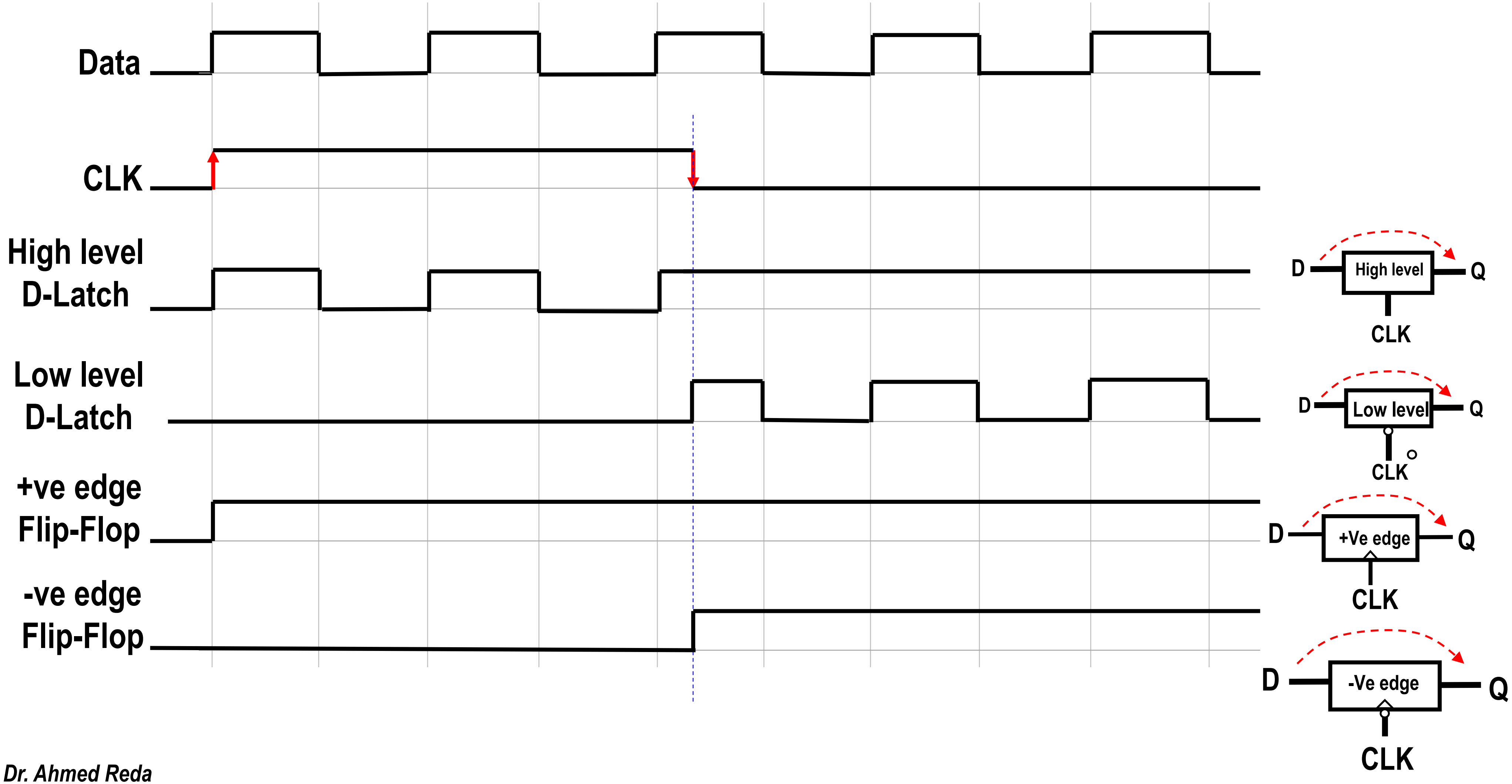
@CLK= ↑, Q=D



@CLK= ↓, Q=D

Memory Cell (2/2)

- ❑ Timing diagram for D-Latch and D-Flip-flop



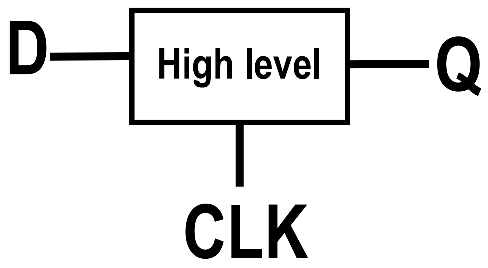
Outlines

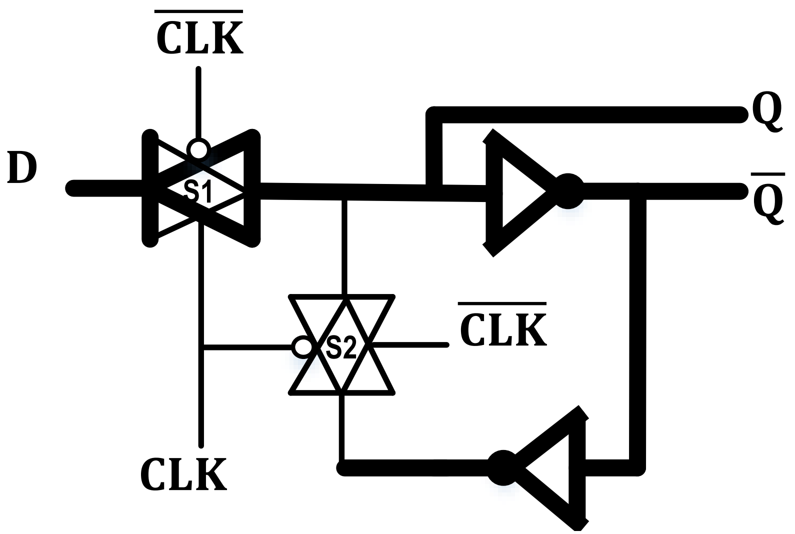
- Memory Cell
- D-Latch
- D-Flip-Flop

D-Latch (1/2)

High level sensitive D-Latch

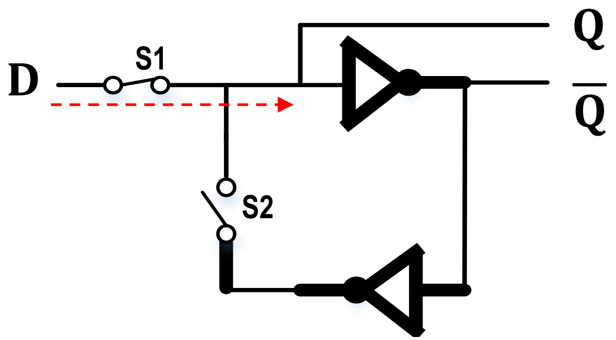
❑ Circuit diagram





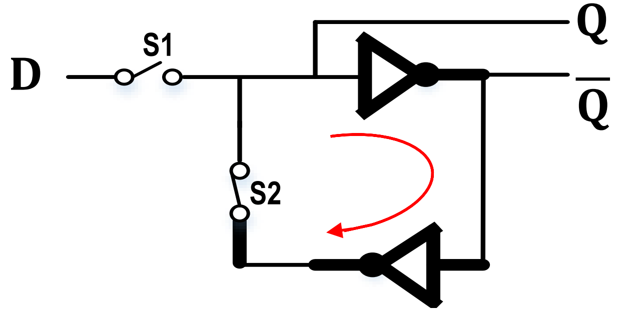
❑ Operation

- ❑ @ CLK= “1”, S1 and S2 are ON and OFF respectively. Q=D



- ❑ @ CLK= “0”, S1 and S2 are OFF and ON respectively. Q=stored D

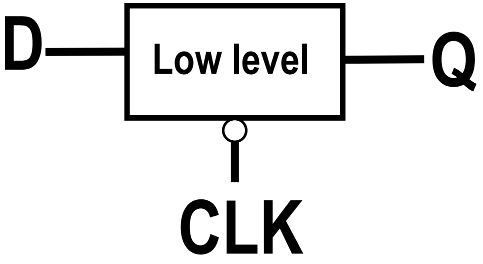
  Data will be circulated inside the cell

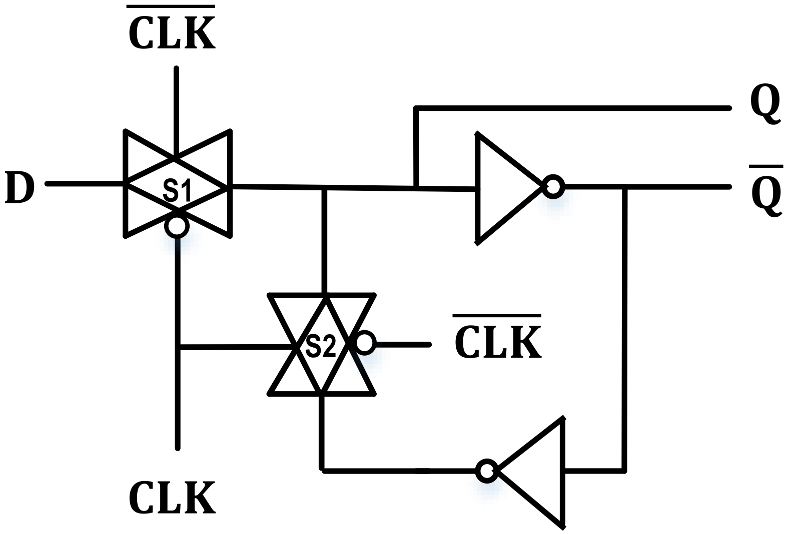


D-Latch (1/2)

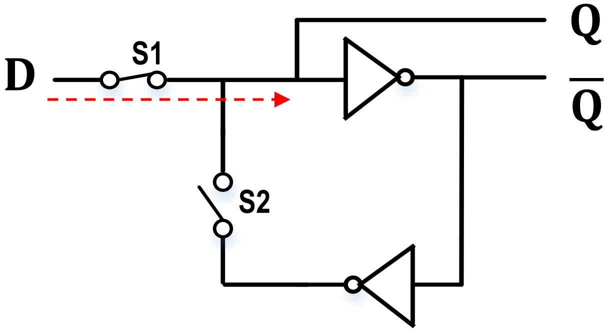
Low level sensitive D-Latch

- ❑ Circuit diagram



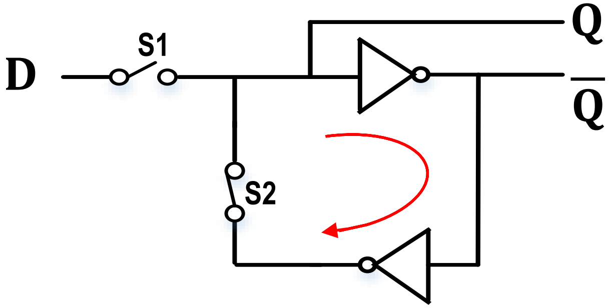


- ❑ Operation
  - ❑ @ CLK= “0”, S1 and S2 are ON and OFF respectively. Q=D



  - ❑ @ CLK= “1”, S1 and S2 are OFF and ON respectively. Q=stored D

    Data will cycle inside the cell

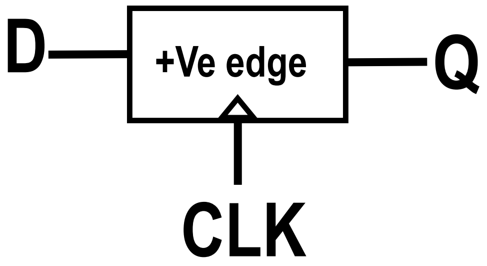


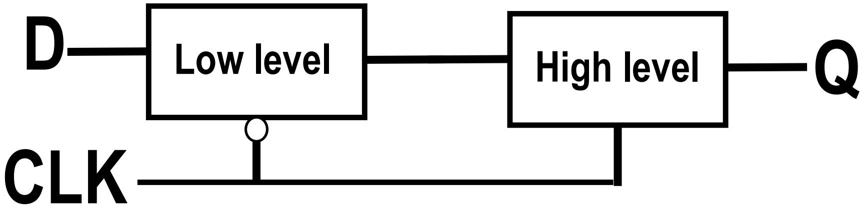
Outlines

- Memory Cell
- D-Latch
- D-Flip-Flop

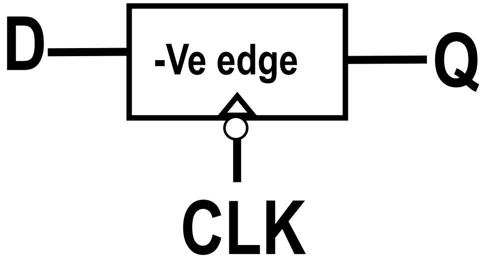
D-Flip-Flop (1/7)

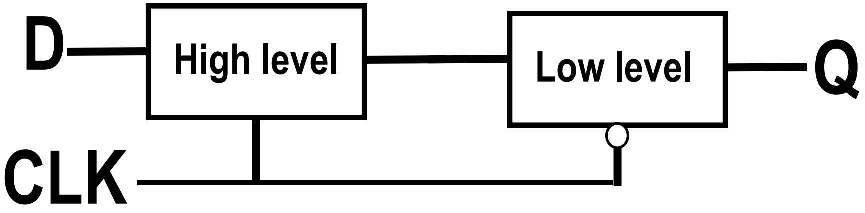
- ❑ It is a combination of two sensitive D-latch "Master and Slave"
- ❑ +ve edge D-Flip Flop consists of Low level D-latch followed by High level D-latch





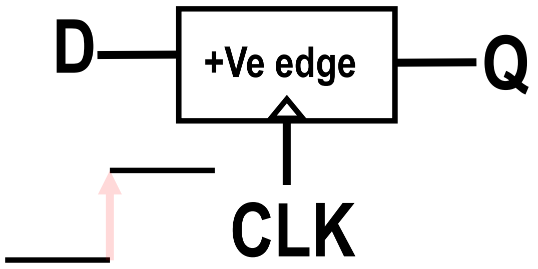
- ❑ -ve edge D-Flip Flop consists of High level D-latch followed by Low level D-latch

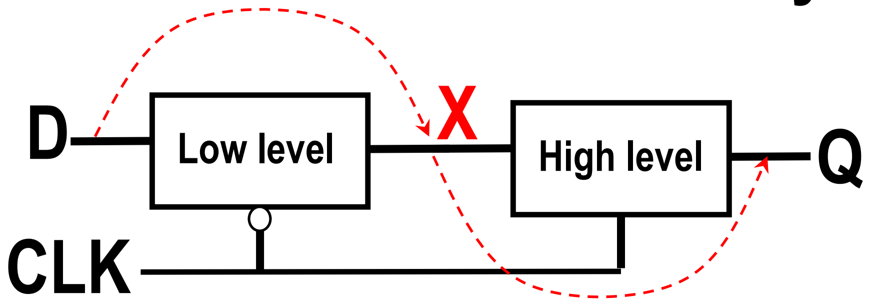


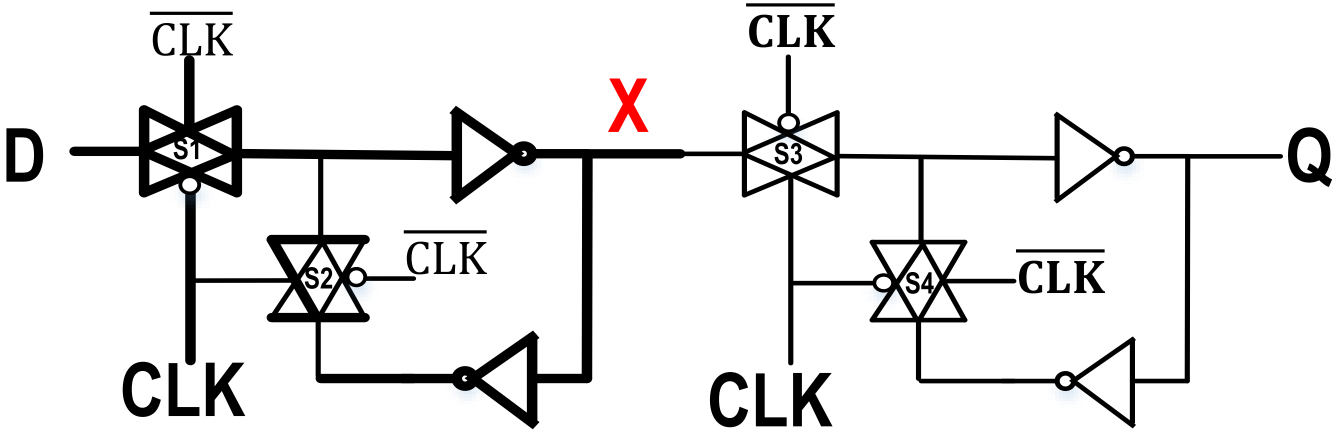


D-Flip-Flop (2/7)

- It is a combination of **two** sensitive D-latch “Master and Slave”
- **+ve edge** D-Flip Flop consists of **Low** level D-latch followed by **High** level D-latch

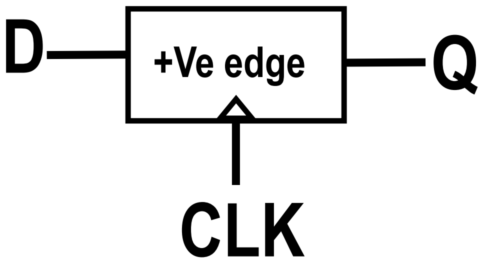


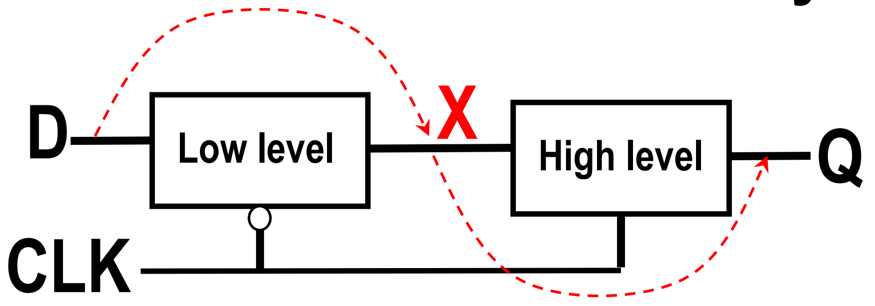


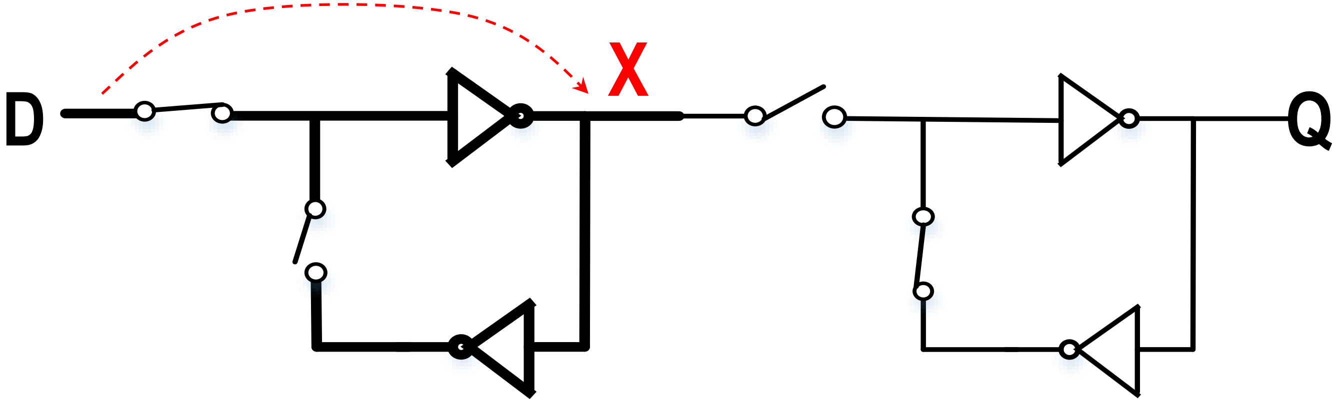


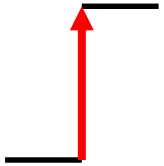
D-Flip-Flop (3/7)

- It is a combination of two sensitive D-latch "Master and Slave"
- +ve edge D-Flip Flop consists of Low level D-latch followed by High level D-latch





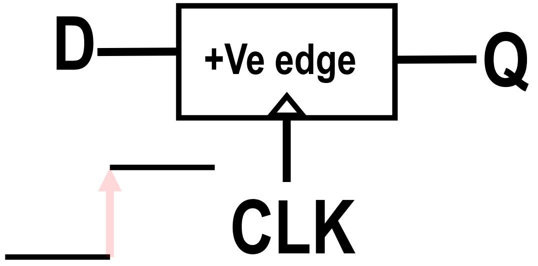


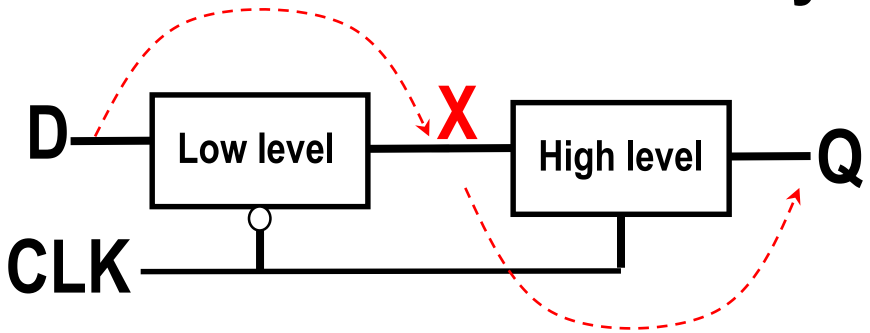


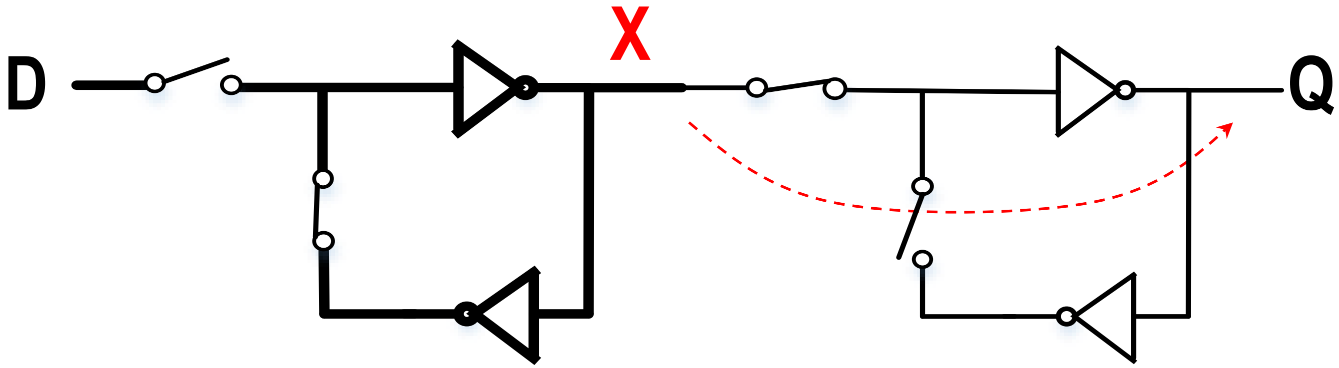
- When CLK="0" Data transfers from D to X=D,

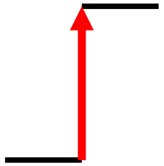
D-Flip-Flop (4/7)

- ❑ It is a combination of two sensitive D-latch “Master and Slave”
- ❑ +ve edge D-Flip Flop consists of Low level D-latch followed by High level D-latch





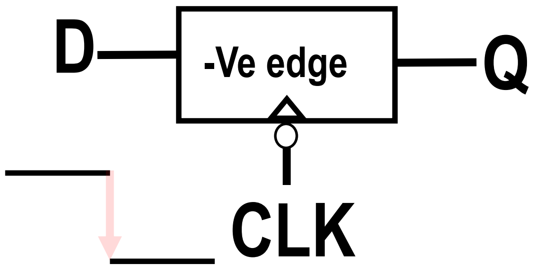


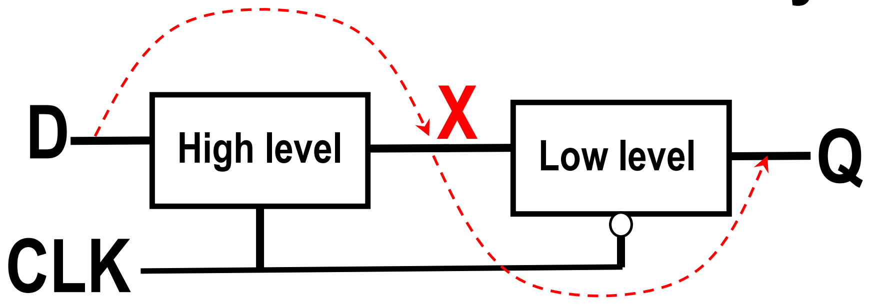


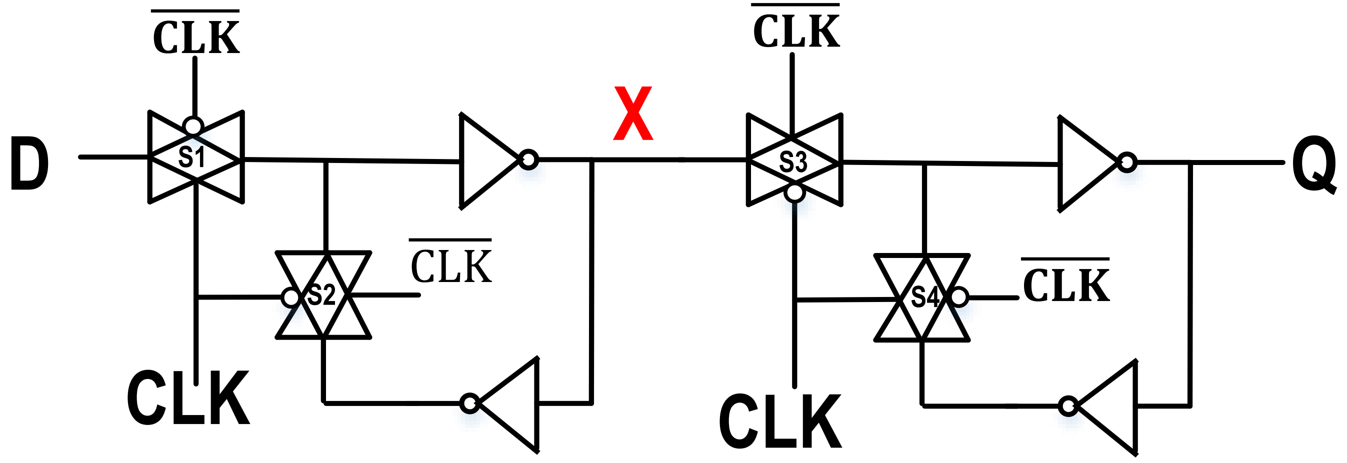
- ❑ When CLK=“0” Data transfers from D to X=D,
- ❑ When CLK=“1” Data transfers from X to Q=D,

D-Flip-Flop (5/7)

- ❑ It is a combination of two sensitive D-latch "Master and Slave"
- ❑ -ve edge D-Flip Flop consists of High level D-latch followed by Low level D-latch

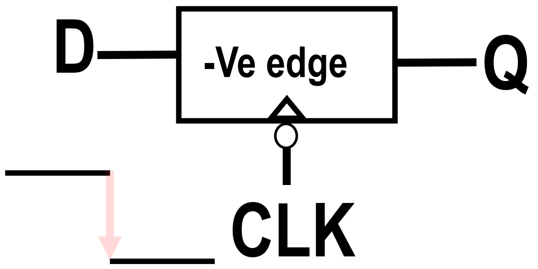


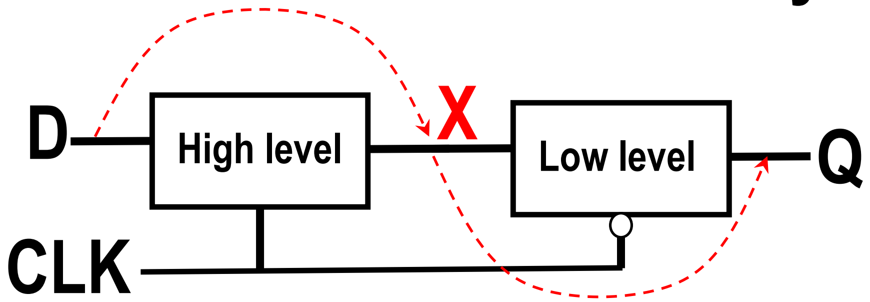


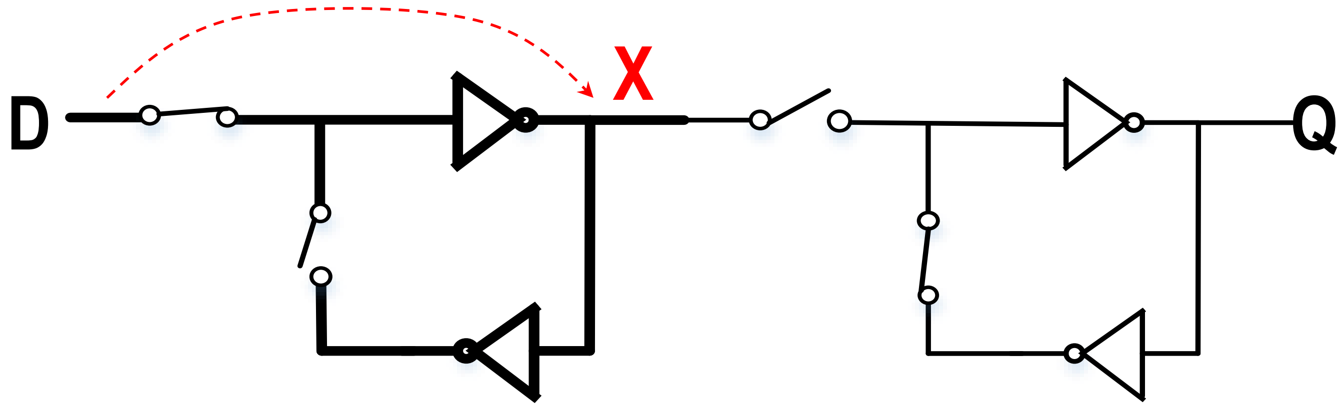


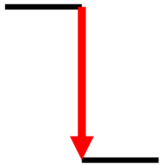
D-Flip-Flop (6/7)

- It is a combination of two sensitive D-latch “Master and Slave”
- -ve edge D-Flip Flop consists of High level D-latch followed by Low level D-latch





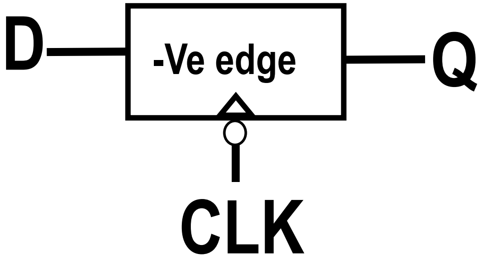


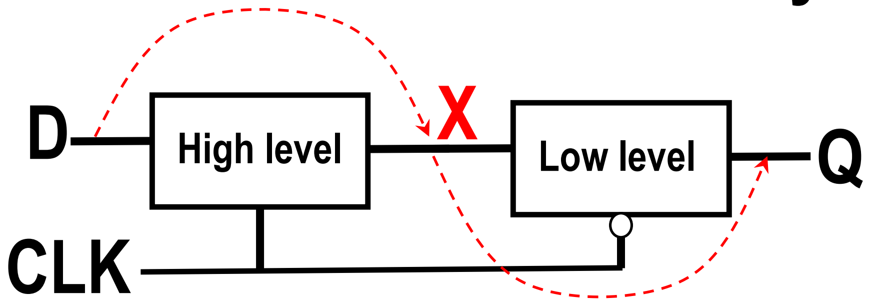


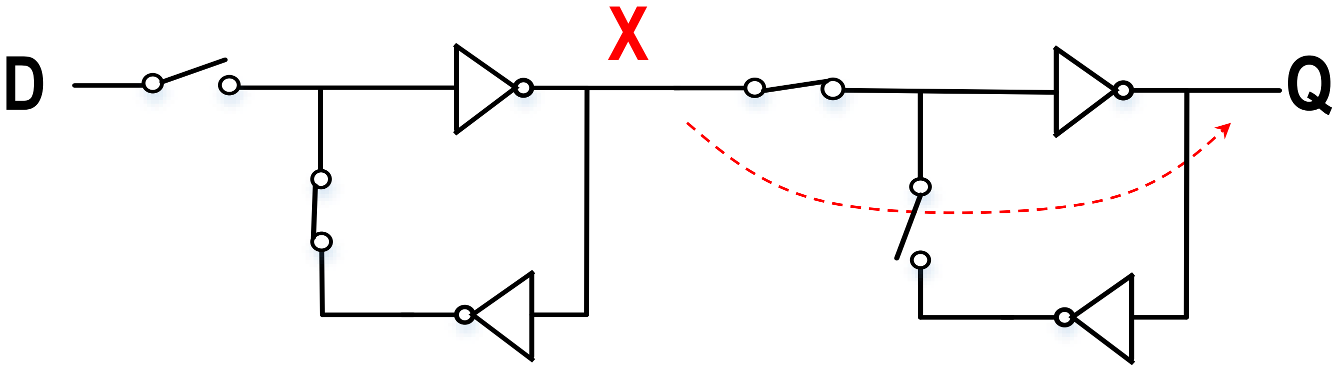
- When CLK=“1” Data transfers from D to X=D,

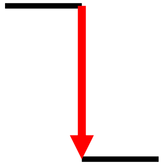
D-Flip-Flop (7/7)

- It is a combination of two sensitive D-latch “Master and Slave”
- -ve edge D-Flip Flop consists of High level D-latch followed by Low level D-latch









- When CLK=“1” Data transfers from D to X=D,
- When CLK=“0” Data transfers from D to X=D,

Thanks!